

Dynamic migration mechanism of borneol synergistic polyethylene glycol for the construction of silicon-acrylate antifouling coating

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ABSTRACT

Silicone-based coatings are vulnerable to marine fouling organisms in static environments, which seriously hampers its practical application. Here, an amphiphilic acrylate polymer is designed by hydrophilic polyethylene glycol and antibacterial borneol, which is introduced into polydimethylsiloxane (PDMS) by covalent linking to prepare antifouling coatings. XPS and CA tests prove that amphiphilic acrylate polymer can migrate to coating surface in the marine environment to maximize antifouling properties. *Mytilus edulis* foot protein adhesion test find it can weaken interface force between Mefp-3 and surface. Laboratory antifouling performance evaluations confirm that the coatings are effective against the adhesion of bacteria (86.72 % reduction for *E. coli*), diatoms (87.76 % reduction for *Nitzschia closterium*). The good antifouling properties of coatings are ascribed to the synergistic effect among unique bicyclic monoterpene structure of natural non-toxic borneol and amphiphilic acrylate polymer on surface of coatings. Moreover, the coating is less harmful to the marine ecological balance than traditional coatings containing toxic biocides due to the introduction of natural non-toxic borneol. This work contributes new insights in the exploration of environmentally friendly coating with excellent static antifouling performance.

1. Introduction

The attachment and reproduction of organisms on the surface of equipment in the aquatic environment is known as marine fouling [1–3]. The accumulation of marine fouling induces significant negative impacts in economy and environment. For example, a large number of marine biofoulings adhere to the vessel can lead to increasing frictional drag and fuel consumption [4,5]. Meanwhile, it is also a source of invasive marine species [6,7]. Various methods have been developed to address marine fouling, such as mechanical cleaning, ultrasonic cleaning and biomimetic antifouling, etc. [1,8]. These methods have been applied in practice, but they are limited by funds and actual conditions. Nowadays, the application of marine antifouling coatings is still an effective strategy to control marine biofouling [9]. However, antifouling coatings containing biocides can cause irreversible effects on organisms in the aquatic environment, such as teratogenicity and growth inhibition [10,11]. Therefore, the exploration of environment protecting and

efficient non-toxic antifouling coatings remains an urgent task.

Silicon-based materials are extensively applied in marine fouling release coating (FRC) on account of low surface energy and low modulus of elasticity [12]. These properties are beneficial to reduce the interaction force between biofoulings and the interface, which can remove biofoulings from the hull surface rely on shear forces [13–16]. In addition, the cross-linked system with excellent chemical stability in seawater environment significantly extends the service life of coatings [17]. However, silicon-based coatings fail to resist the settlement of biofoulings in static environments, thus limiting the practical application [18–20]. In order to solve the above defects, silicon-based coatings have been modified by introducing natural antifouling agents, amphiphilic substances or increasing crosslinking density [21–25].

Borneol is a naturally substance extracted from dipterocarp and peppermint that has attracted extensive attention in recent years. It has broad-spectrum antibacterial properties due to its unique bicyclic monoterpene structure [26–29], and it is less toxic and environmentally

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friendly. Song et al. confirmed that the introduction of natural borneol in the acrylate coating can effectively inhibit bacterial adhesion. However, the introduction of borneol into silicon-based coatings has been less reported [30]. Borneol is easy to enrich in non-surface layer if combine with silicon-based polymers. Polyethylene glycol (PEG) with highly hydrophilic can migrate to the surface of the coatings to form a hydrated layer and block the adhesion of proteins/bacteria in the aquatic environment [31–34]. Therefore, amphiphilic substance is constructed by introducing hydrophilic and hydrophobic may be effective anti-fouling strategy in silicon-based coatings. As far as we know, study on the synergy of borneol and PEG to improve static antifouling of silicon-based coatings have rarely been reported.

In this work, we successfully synthesize an amphiphilic acrylate polymer with polyethylene glycol side chain and borneol structure by radical polymerization, it is introduced into PDMS by condensation reaction for solving static antifouling. In the marine environment, PEG chains pull borneol group to surface of coatings due to its hydrophilic nature, improving the ability of coatings to resist biofoulings. *Mytilus edulis* foot protein adhesion test confirm that amphiphilic acrylate polymer can weaken the interaction force between the Mefp-3 and coating surface. In addition, the coating is covalent connections to form cross-linked networks by using the silane crosslinker, improving mechanical properties and stability of the coatings. The coating provides a strategy for the development of environmentally friendly static marine antifouling coatings in practical applications.

2. Experimental section

2.1. Synthesis of amphiphilic acrylate polymer (PABG)

PABG polymer was synthesized by free radical polymerization as shown in Fig. 1a. Typically, xylene (40 g) and PGM (35 g) were introduced to a three-necked flask equipped with nitrogen and condensing equipment, stirred continuously and heated to 85–90 °C. Then, MMA (31.5 g), BA (16 g), 2-HEA (5 g), PEGMA (7 g), PGM (20 g), IBOA (10.5

g), AIBN (0.3 g), AIMN (0.4 g) and 2, 4-Diphenyl-4-methyl-1-pentene (0.2 g) were stirred to form mixed solution. The mixed solution was added dropwise to three-necked flask and stirred continuously for 4 h. Then, BPO (0.5 g) was dissolved in xylene (20 g) and added to the above mixed solution dropwise 0.5 h. The PABG was obtained after holding for 2 h at the end of drip adding.

2.2. Preparation of PDMS-Px coatings

PDMS-Px coatings were prepared by condensation reaction in Fig. 1b. Briefly, PABG with different mass percentages were dissolved in xylene, 6 g HTPDMS, 150 μ l DBTDL and 0.3 g METES were added to above solution to form a homogeneous solution. Then, the mixture was coated on the treated slides and left for 24 h for subsequent experiments. The feeding ratio were detailed shown in Table S1.

3. Results and discussion

3.1. Structural characterization of PABG polymer and PDMS-Px coatings

The molecular structure of PABG polymer is determined by ^1H NMR and FT-IR tests. As shown in Fig. 2a, the peak at 3.59 ppm is ascribed to the $-\text{CH}_3$ of MMA. In addition, peak at 3.75 ppm is assigned to the $-\text{CH}_3$ absorption of BA and 2-HEA, and the peak at 4.46 ppm correspond to the IBOA. The results indicate that PABG polymer is synthesized successfully. Furthermore, the FT-IR spectra of PABG polymer and 2-HEA are shown in the Fig. 2b. The characteristic absorption peak at 3537 cm^{-1} is attributed to O–H stretching vibrations, the bands at 2800 cm^{-1} – 3000 cm^{-1} are ascribed to the bending vibrations of CH_3 and CH_2 , and the peak located at 1720 cm^{-1} is associated with stretching vibrations of C=O. The characteristic absorption peak of C=C at 1640 cm^{-1} disappears for PABG polymer, indicating the completion of the monomer reaction. The functional groups of the samples are tested by ATR-FTIR. As shown in Fig. 2c, the ATR-FTIR spectra of PDMS and PDMS-P23 show sharp peaks at 1010 cm^{-1} and 1080 cm^{-1} , which are

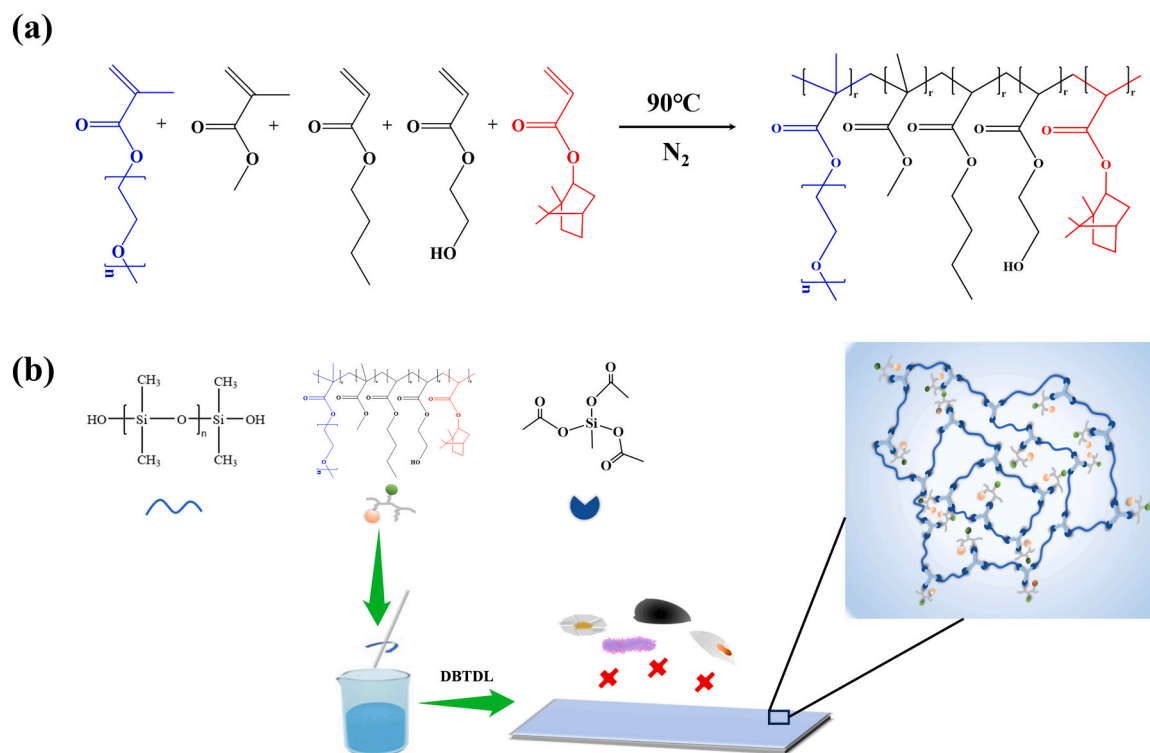


Fig. 1. (a) Synthesis route of the PABG polymer. (b) Schematic diagram of the synthesis of PDMS-Px coatings.

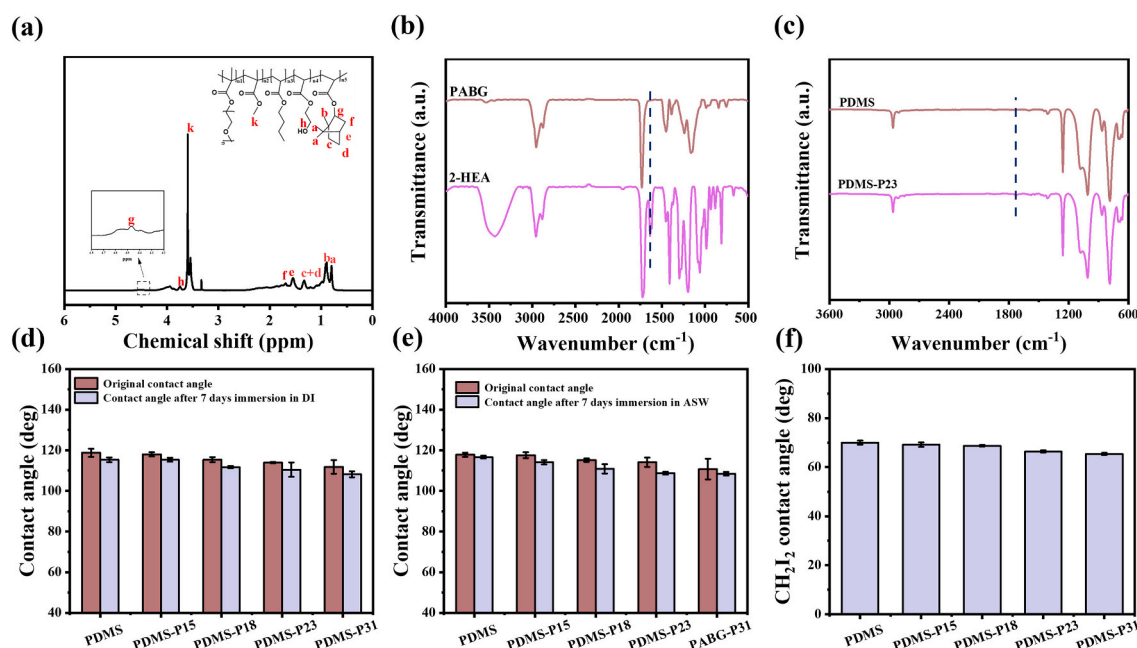


Fig. 2. (a) ¹H NMR spectrum of the PABG polymer. (b) FT-IR spectra of PABG polymer and 2-HEA. (c) ATR-FTIR spectra of PDMS, PDMS-P23 coatings. (d) DI water and (e) ASW CA of PDMS-Px coatings before and after immersion in ASW for 7 days. (f) Surface energy of the PDMS-Px coatings after immersed in ASW for 7 days.

attributed to the Si—O—Si stretching vibration in PDMS. In addition, the C=O characteristic peak of PABG is observed at 1720 cm⁻¹. No characteristic absorption peak of —OH is found in the PDMS-P23 coating spectra, indicating the successful cross-linking of PABG with HTPDMS. Gel permeation chromatography (GPC) test illustrates that the molecular weight (Mw) of the PABG is 1.85×10^4 g/mol and the polydispersity index (PDI) is 1.55. The PDI of polymer is low, which indicating a homogeneous molecular weight.

3.2. Surface property of PABG in PDMS-Px coatings

The migration behavior of amphiphilic substances in the coating is studied by testing the changes of contact angle (CA), roughness and elemental content before and after the coating is immersed in seawater. The change in CA of PDMS-Px coatings before and after 7 days immersion in ASW is tested. As shown in Fig. 2d, PDMS exhibits the largest CA of 118.53°. The CA of PDMS-Px coatings gradually decrease with the increase of PABG content, and the CA of PDMS-P31 coating reach 111.8°, which is caused by the hydrophilic properties of PEG. After 7 days of ASW immersion, the CA of the coatings further reduce due to the hydrophilic PEG segment contained in the PDMS-Px coating. As shown in Fig. 2e, the deionized water (DI) is replaced with ASW and the trend is similar to the ASW test. In general, the adhesion ability of biofouling is related to wettability, and it has been studied that hydrophilic surface can form a strong hydration layer to block the adhesion of biofouling [35,36]. Additionally, CA tests are performed at non-polar CH₂I₂ and the surface energy of the PDMS-Px coatings is calculated (Fig. S1). As shown in Fig. 2f, the surface energy of PDMS-Px coatings reach a low surface energy range, which facilitate the removal of fouling organisms from the coating surface.

Scanning electron microscopy (SEM) is carried out to prove the stability of coatings under marine environment. The surface topography of the PDMS-Px coatings before and after 7 days of ASW immersion is investigated, and it can be seen from Fig. S2a and b that ASW immersion have less affect the flatness of the coatings, proving that the PDMS-Px coatings are stable in the marine environment. The surface roughness of PDMS-Px coatings is tested by using atomic force microscopy (AFM). Fig. S3a and b are the AFM images of PDMS-Px coatings before and after 7 days of immersion in ASW, and the value of Rq and Ra are used to

express the roughness. The Rq and Ra of PDMS coating are 2.70 nm and 1.39 nm before immersion in ASW, and the introduction of a small amount of PABG polymer increase the roughness of the coatings. However, the roughness decreases with the substantial increase of PABG polymer proportion, the value of Rq and Ra are 0.697 nm and 0.548 nm. In addition, the roughness of PDMS vary significantly after immersing in ASW, and the Rq and Ra are 27.9 nm and 14.7 nm. The coatings containing PABG polymer have little change in roughness, the Rq and Ra of PDMS-P31 coating are 0.703 nm and 0.526 nm. The smooth surface is less susceptible to adhesion by fouling, while the rough surface provides attachment sites for fouling and it is susceptible to attachment by fouling [37,38].

High resolution XPS is measured to prove the presence and content of elements on the surface of coatings, the XPS spectrum of all coatings are shown in Fig. S4. Taking PDMS-P31 coating as an example, the binding energy of C—C, C—Si and C—O are 284.6 eV, 285 eV and 286.2 eV (Fig. S4 e), showing that coatings contain C, O and Si elements. Amphiphilic substances migration behavior is further illustrated by coating surface C/Si of original and immersion coatings. As shown in Table 1, the C/Si of the PDMS-Px coatings increase after immersion in seawater. Especially, the C/Si of the surface of PDMS-P23 coating is 2.27 under the original condition, and the C/Si increase to 3.05 after 7 days of immersion in ASW. The results show that seawater immersion cause the PABG to migrate upward in the crosslinked network, while the Si-containing segments shift downward [39].

3.3. Adhesion strength experiment

The adhesion strength of the coatings is key indicator for practical application, and the pull-off method is used to evaluate the adhesion

Table 1

C/Si before and after PDMS-Px coating immersion in ASW for 7 days.

Sample	C/Si	C/Si*
PDMS-P15	2.03	2.19
PDMS-P18	2.16	2.41
PDMS-P23	2.27	3.05
PDMS-P31	2.13	2.41

strength of the coatings. As shown in Fig. S6, the adhesion strength of PDMS is 0.72 MPa, which cannot meet the actual application. After the introduction of PABG, the adhesion strength of PDMS-P31 coating increases to 1.38 MPa. On the one hand, the adhesion strength of PABG is better than PDMS because the hydroxyl groups of the side chains can bond with the substrate. On the other hand, the combination of the PDMS and PABG increase the tensile strength (Fig. S5) of the coatings and increase the adhesion between the inside of the coatings.

3.4. Antibacterial activity

In the marine environment, bacterial adhesion is a critical step in biofilm formation and provides favorable condition for subsequent colonization of organisms. Borneol is a natural antibacterial substance, the inhibition zone experiment is utilized to investigate the antibacterial effect [27,40–43]. The inhibition zone of isoborneol on *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) are indicated in Fig. S8 a and b, respectively. The bacteria inhibition zone can be clearly found, which proves the inhibition ability of isoborneol. To verify the antibacterial effect of coatings, inhibition zone experiments are performed on IBOA, PDMS and PDMS-P31 coating by using *S. aureus* (Fig. S8 c). The surface of IBOA and PDMS-P31 coating is still clean after 48 h, while many bacteria grow at bottom and around the PDMS sample, manifesting that the addition of borneol could effectively avoid the invasion of bacteria. To further demonstrate the antibacterial ability of coatings, *E. coli*, *S. aureus* and *Pseudomonas Xiamenensis* are selected as the model strains and co-cultured with PDMS-Px coatings. The antibacterial ability of the coatings is evaluated by cell counting. The optical photographs of the colonies of *E. coli*, *S. aureus* and *Pseudomonas Xiamenensis* are shown in Fig. 3a, b and S7, it is clearly observed that PDMS shows more colonies than the PDMS-Px coatings, indicating that static antifouling ability of PDMS is poor [44]. From Fig. 3c–e, the PDMS-P23 coating shows the best antibacterial ability with 85 % against *E. coli*, 86.72 %

against *S. aureus* and 76.6 % against *Pseudomonas Xiamenensis*. This indicates that the borneol group in PABG polymer plays a crucial role to resist the attachment of bacteria in the marine environment and decrease the attachment of biofouling.

3.5. *Mytilus edulis* foot protein adhesion test

Mussel edulis foot protein (Mefp) is a class of sticky proteins synthesized by the foot glands of mussels and secrete to the foot filament, including Mefp1–6, preCoID and preCoING. The Mefp-3 is mainly located in adhesive plaques of mussels, which plays an important role in surface adhesion, enabling the mussel foot threads to adhere quickly to the substrate surface underwater [45–47]. Therefore, Mefp-3 protein is chosen to test the interface adhesion between mussel protein and the coating surface. Fig. 4a shows the experimental procedure of Mefp-3 protein adhesion coating with aqueous environment by colloidal probe atomic force microscopy (AFM). Fig. 4b shows the adhesion of Mefp-3 to the blank sample, and it can be seen that the adhesion force of the protein to the interface is approximately 32.9 nN. Fig. 4c shows that the adhesion force between Mefp-3 to PDMS-P15 coating is about 7.32 nN, and Fig. 4d shows that most of the adhesion test values are between 0 and 10 nN, with only a few data with larger values. In summary, the adhesion of mussel proteins to the PDMS-P15 coating is relatively weak, and it can be speculated that the adhesion between Mefp-3 and the surface is significantly reduced due to the upward migration of the PEG fragments in the aqueous environment. We can speculate that the attached mussels are more likely to detach from the sample surface than the PDMS surface under the shear force of the water flow.

3.6. Anti-diatom assay

Halaphora. sp and *Nitzschia closterium* are selected to evaluate the antifouling properties of PDMS-Px coatings. After the diatom

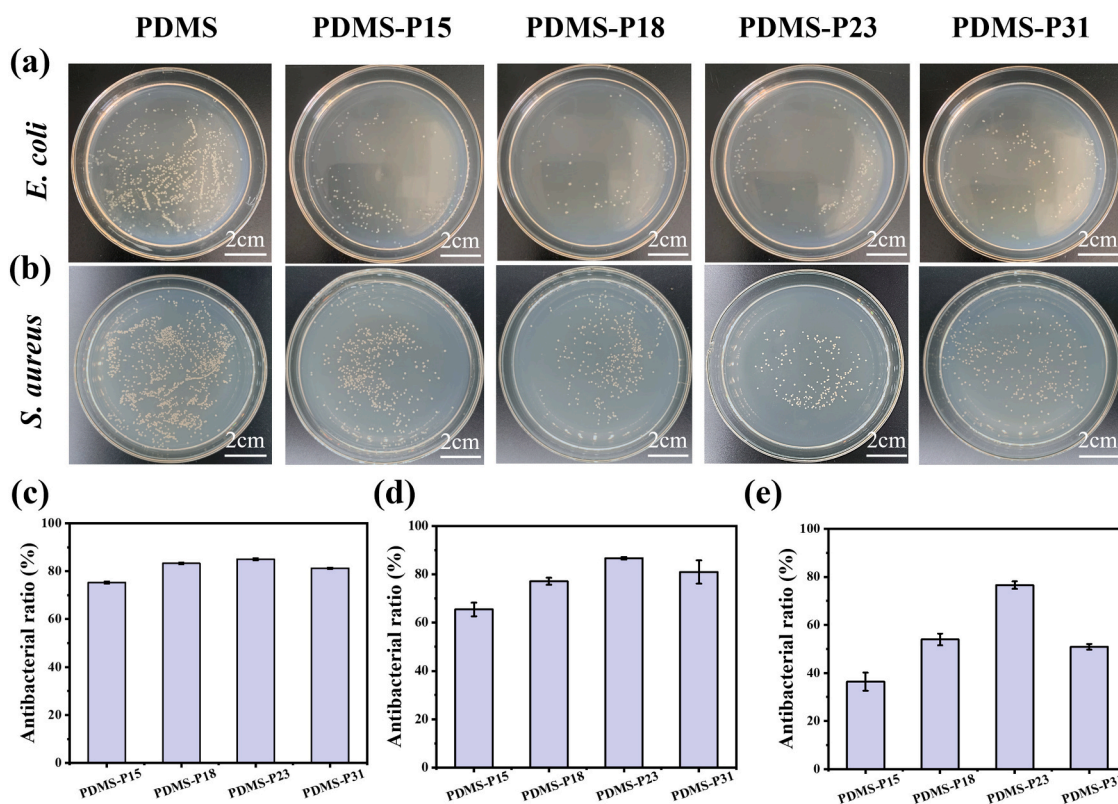


Fig. 3. Optical photographs of (a) *E. coli* and (b) *S. aureus* colonies after co-culture with PDMS-Px coatings. (c) Antibacterial ratio of *E. coli* colony on PDMS-Px coatings. (d) Antibacterial ratio of *S. aureus* colony on PDMS-Px coatings. (e) Antibacterial ratio of *Pseudomonas Xiamenensis* colony on PDMS-Px coatings.

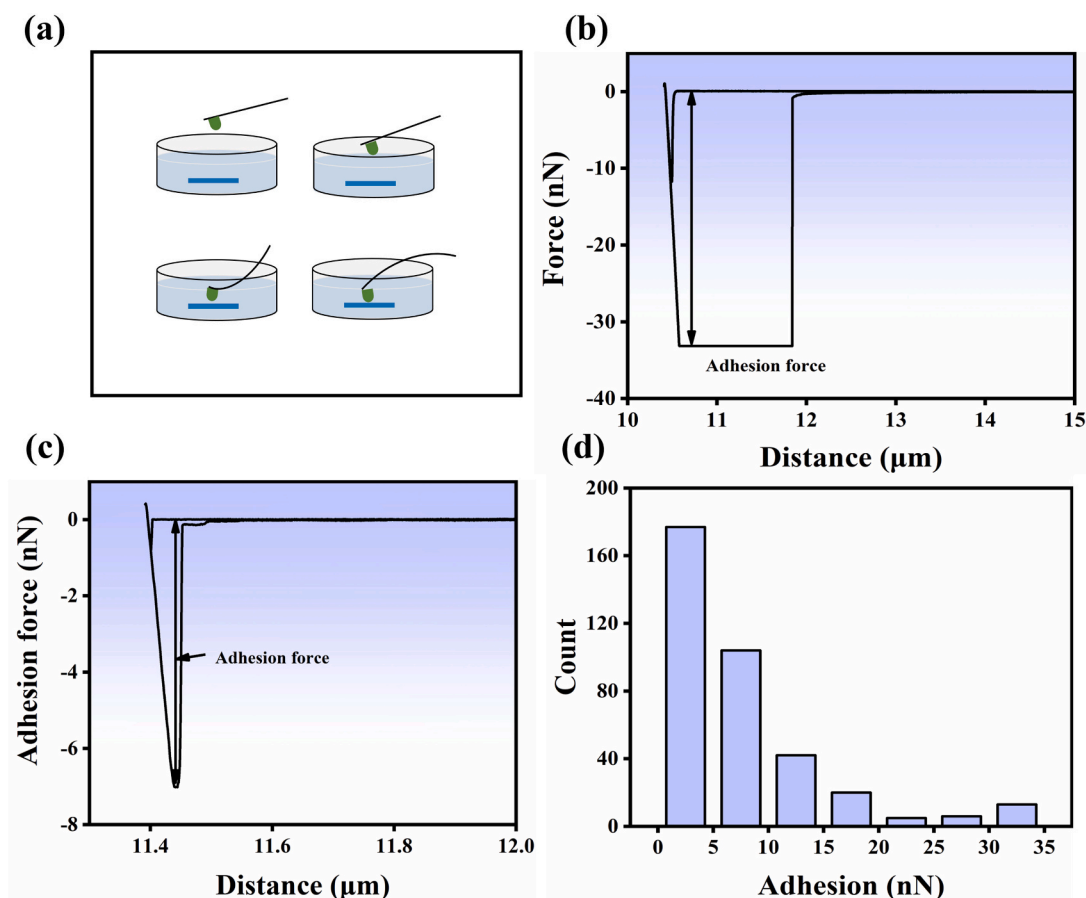


Fig. 4. (a) Schematic of the colloidal probe atomic force microscope experimental. (b) Mefp-3 to PDMS distance-adhesion curve. (c) Mefp-3 to PDMS-P15 coating distance-adhesion curve. (d) Distribution range of Mefp-3 and PDMS-P15 coating interfacial adhesion test values.

experiment, coatings surface is rinsed with a 30 Pa water stream to observe its release property. As shown in Fig. 5a, many diatoms attach to the surface of PDMS samples, while the number of attached diatoms is significantly reduced under static conditions with the introduction of PABG polymer. And the anti-diatom ratio of the PDMS-P23 coating approach 80 % (Fig. 5c). The adhesion number of diatoms on surface of PDMS-Px coatings decreases after rinsing the coating surface with a 30 Pa water stream (Fig. 5b), this is mainly due to the flexible backbone and low surface energy of PDMS reduce the adhesion strength of fouling organisms to the coatings surface. However, the release ratio of PDMS-Px coatings decrease with the increase of PABG polymer content from 65.7 % to 21.6 % (Fig. 5d). The static antifouling ability is improved due to the increase of PABG content, thus there is little difference compare with the cell number of surface diatoms before the water flow washes. In addition, the static antifouling ability of the coatings is tested by the *Nitzschia closterium*. As shown in Fig. 6a-e, the adhesion diatoms cell numbers on the surface of the PDMS-Px coatings are significantly less than that of PDMS, and the anti-diatom trend is similar to *Halimphora. sp.*, where the anti-diatom ratio of PDMS-P23 coating is 87.76 %. XDLVO theory is applied to calculate the interaction energy between coatings and diatom [48–50]. As shown in Fig. S9a and b, U^{EL} and U^{AB} play a major role for short-range adhesion (<1.4 nm), and U^{XDLVO} of PDMS and PDMS-P23 coating is positive value indicating that there is repulsive force between the diatom and coatings. It can be seen from Fig. S9c, the U^{XDLVO} of the PDMS-P23 coating is larger than PDMS, indicating that the PDMS-P23 coating repels the adhesion of diatom more than PDMS due to PDMS-P23 coating contains PEG side chain [51], resulting in a more negative potential than PDMS. In addition, the diatom cells are easy to adhere to the surface of coatings at long-distance. However, considering the low energy property of coatings and hydrodynamic condition of

ocean [16], diatom cells are easily separated from the coatings.

3.7. Pseudo-barnacle adhesion strength experiment

Barnacles are one of the common types of macro-organisms, pseudo-barnacle adhesion strength experiments are performed according to ASTM D5618–94 (2011) to explore the adhesion strength of biofoulings to interfaces. As shown in Fig. 5d, the removal strength of PDMS-Px coatings is less than 0.15 MPa, while the adhesion strength of PDMS is 0.23 MPa, which is significantly higher than that of PDMS-Px coatings. Due to the flexible chain of PDMS, which has the characteristics of low surface energy and low modulus of elasticity. The crosslinking of PABG polymer and PDMS reduces the adhesion strength of fouling organisms to the interface and remove them under the action of water flow. Diatom desorption experiments prove that most fouling organisms can be shed from the surface after a certain force is applied.

3.8. Mussel multichoice assay and settlement

Mussels are also common macrofouling that attach to underwater facilities. In this paper, mussels are selected to assess the resistance of coatings to macro-organism. The glass plates, PDMS and PDMS-Px coatings are randomly distributed on the epoxy board, and 18 mussels are randomly placed on the coatings. As shown in Fig. 7a, b and c, most mussels attempt to find a suitable surface for adhesion. After 48 h, plaques secreted by mussels on the coatings are shown in Fig. 7d, e and f. Mussels are mostly attached to the glass slides and PDMS, and only a few plaques are observed on the sample surface. Furthermore, the adhesion strength between coatings and mussel is weaker than that of glass plates (Fig. S10). During the experiment, we find that the mussel individuals

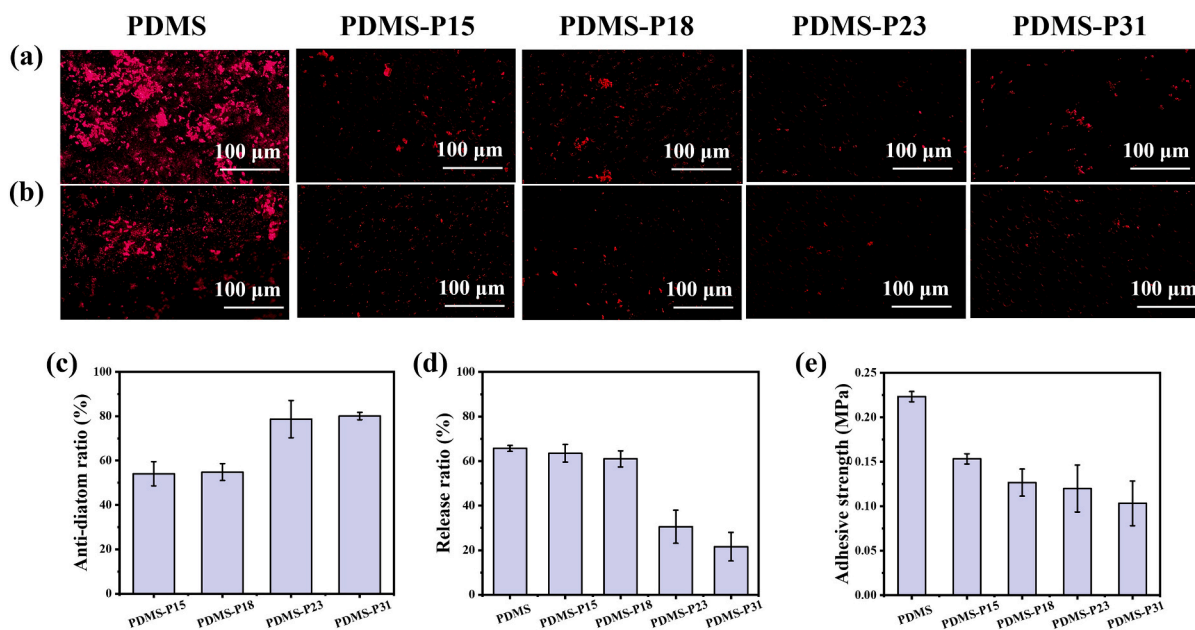


Fig. 5. Anti-diatom (*Halamphora. sp*) property of PDMS-Px coatings. (a) Fluorescence image of PDMS-Px coatings in static environment for 48 h. (b) Fluorescence image of the PDMS-Px coatings after washing in a 30 Pa water jet. (c) Laboratory anti-diatom experiment of PDMS-Px coatings under static conditions. (d) Diatom release ratio of PDMS-Px coatings under dynamic conditions (30 Pa water flow). (e) Removal strength of adhere pseudo-barnacle from PDMS-Px coatings.

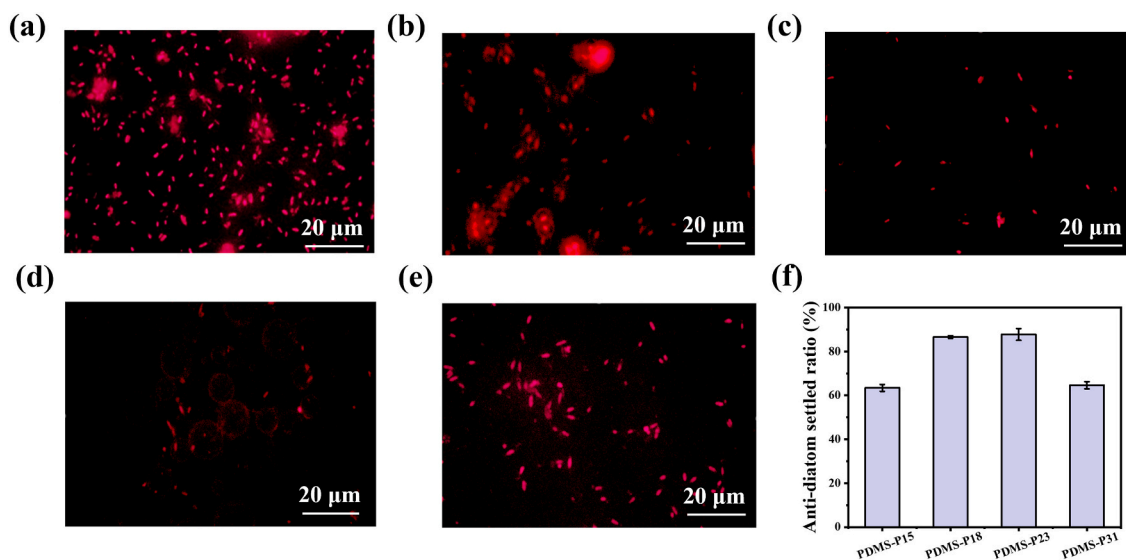


Fig. 6. Anti-diatom (*Nitzschia closterium*) property of PDMS-Px coatings. Fluorescence image of (a) PDMS, (b) PDMS-P15, (c) PDMS-P18, (d) PDMS-P23, (e) PDMS-P31 coatings immersed in algae solution for 7 days. (f) The diatoms inhibition ratio of PDMS-Px coatings.

tend to adhere to each other, and Fig. 7g show that only 8.1 % of the mussels choose to adhere to the sample surface, this may be due to the avoidance effect of borneol of in the PABG polymer on mussels, which indicating that PDMS-Px coatings have good mussel adhesion resistance ability.

4. Conclusion

In summary, an amphiphilic acrylic polymer-modified polydimethylsiloxane coatings are successful prepared for efficient static marine antifouling. The introduction of PEG chains and natural borneol in PABG polymer blocks adhesion of biofouling, and weaken interface force between surface of coatings and biofouling. The inhibition ratio of PDMS-P23 coating against *E. coli*, *S. aureus* and *Pseudomonas Xiamenensis*

reach to 85 %, 86.72 % and 76.6 %. *Mytilus edulis* foot protein adhesion test confirm that PABG can weaken the mussel protein and interface forces. Moreover, the diatom inhibition ratio of PDMS-P23 coating is approach to 80 % for *Halamphora. sp* and 87.76 % for *Nitzschia Closterium*. PDMS contributes significantly to fouling release on account of its low surface energy, and the release rate of PDMS-Px coatings are greater than 60 %, pseudo-barnacle removal strength of the coatings are less than 0.15 MPa. Furthermore, PDMS-Px coatings exhibit excellent stability and environmentally friendly. This work provides a promising strategy for the development of environmentally static marine anti-fouling coatings.

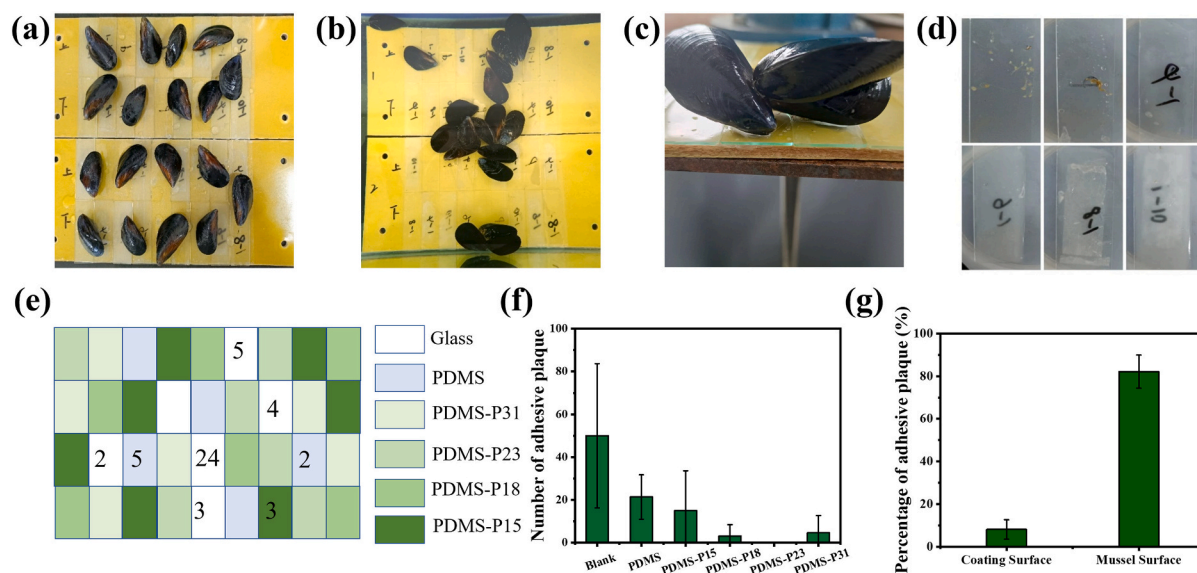


Fig. 7. Mussel multichoice assay and settlement results of PDMS-Px coatings. (a) Random distribution of mussels on the surface of coatings. (b) Image of the position after 48 h. (c) Photographs of the mussels adhered to the surface. (d) Photographs of plaques on PDMS-Px and PDMS coatings. (e) The number of plaques adhered to different surface in one experiment. (f) The average number of the plaques secreted by the mussels on each sample. (g) Percentage of plaques on samples and on mussel individuals.

CRediT authorship contribution statement

Wei Zhao: Conceptualization, Methodology, Investigation, Software, Writing – original draft. **Rongrong Chen:** Supervision, Funding acquisition, Writing - review & editing. **Peili Liu:** Supervision. **Jing Yu:** Data curation, Resources. **Qi Liu:** Supervision. **Jingyuan Liu:** Data curation, Resources. **Jiahui Zhu:** Resources. **Shifeng Guo:** Funding acquisition, Data curation. **Jun Wang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2023.107946>.

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